10PB24SCI04

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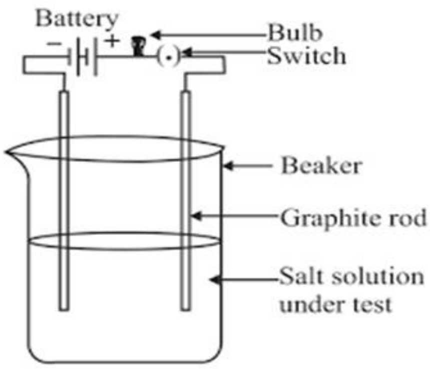
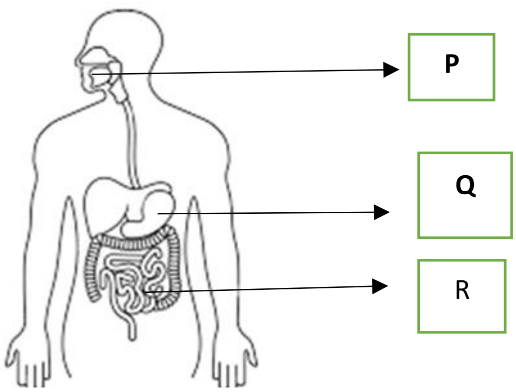
CLASS X **SCIENCE 086** **MM80** **Time 3h**

General Instructions:

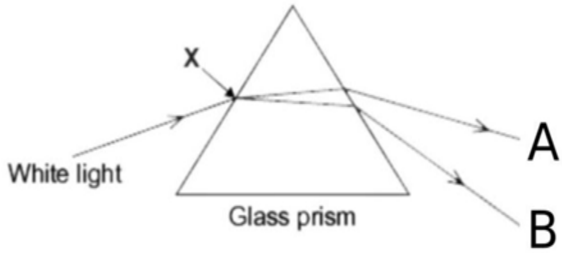
1.All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
2.Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
3.Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
4.Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
5.Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
6.Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1/2/3 marks.

	Section-A Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.	
1	Identify 'x', 'y' and 'z' in the following balanced reaction. x Fe (s) + y H_2O (g) Fe_3O_4 (s)+ z H_2 (g) A. 2,3,4 B. 2,4,2 C.3,4,4 D.4,2,3	1
2	Anju took sample P(salt solution) and sample Q(solid salt) , to test the conductivity and made some conclusions. Which of the following is the correct match from the table?  Sample P/Q Sample / Observation / Conclusion P / a. Bulb glows / c. No movement of ions Q / b. Bulb does not glow / d. Free movement of ions A . P - a - d B. Q - b – d C. P - b - c D. Q - a - c	1
3	Vani collected some of the samples of substances for the experiment in Chemistry lab. Which of the following turns blue Litmus Paper into red? M. toothpaste N. orange juice O. common salt P. milk of magnesia Q. baking soda A. M,P,Q B. N & O C. M,O,Q D. Only N	1
4	An aqueous solution 'A' turns the phenolphthalein solution pink. On addition of an aqueous solution 'B' to 'A', the pink colour disappears. Which of the following statement is true for the solutions 'A' and 'B'.	1

	A. A is strongly basic and B is a weak base. B. A is strongly acidic and B is a weak acid. C. A has a pH greater than 7 and B has a pH less than 7. D. A has a pH less than 7 and B has a pH greater than 7.	
5	When hydrogen sulphide gas is passed through a blue solution of copper sulphate, a black precipitate of copper sulphide is obtained and the sulphuric acid so formed remains in the solution. This reaction is an example of A. Combination reaction B. Decomposition reaction B. Displacement reaction D. Double displacement reaction	1
6	Calcium and magnesium start floating after reaction with water because A. bubbles of hydrogen gas formed stick to the surface of the metal B. hydrogen gas evolved catches fire C. reaction is highly endothermic D. reaction is least vigorous	1
7	Which of the following is an endothermic reaction? A. Burning of candle. B. Cooking of food. C. Decomposition of Vegetable matter. D. Reaction of Sodium with air	1
8	Atul suffers from muscle cramp resulting from his strongest physical workout . Which biochemical process leads to this discomfort? A. generation of lactic acid B. production of Acetic Acid C. formation of pyruvic acid D. creation of hydrochloric acid	1
9	During which of the following stages of the circulation of blood in a normal human being, the oxygenated blood is pumped to all parts of the body? A. contraction of the left atrium B. contraction of left C. relaxation of the right atrium D. relaxation of the right ventricle	1
10	Consider the diagram provided below. Which of the following nutrients are digested in the organs P,Q and R  A. P fat, Q starch, R lactose B. P lactose, Q protein, R starch C. P lactose, Q fat, R starch D. P starch, Q protein, R lactose	1
11	There was a cerebellar dysfunction in a patient. Which of the following activities will get disturbed in this patient as a result of this? A.Salivation B.Hunger control C.Posture and balance D.Regulation of blood pressure	1
12	Sex determination depends upon the environment in A.Birds B.Amphibians C.Reptiles C.Fishes	1
13	A concave mirror produces magnification (m)= -2.What would be the interpretation of the given value of m?	1

I .the image is real and inverted
II. the image is virtual and erect
III. the image is diminished
IV. the image is two times enlarged
A.Only II   B.II and III   C.I and III   D.I and IV

14 (1)



Choose the correct option for the colour of rays for A and B.

	Colour of Ray A	Colour of Ray B
A.	Violet	Red
B.	Green	Yellow
C.	Red	Violet
D.	Violet	Indigo

15 If the total energy at the trophic level of producers in an ecosystem is ‘E’, then which
of the following corresponds to the energy available to the tertiary consumers? (1)
A.E/10   B.10 X E   C.E/1000   D.1000 X E

16 What is the consequence of biomagnification in environmental science? (1)
A. decreasing soil fertility   B. accumulating toxins in food chains
C. increasing atmospheric temperature   D. enhancing species diversity

Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

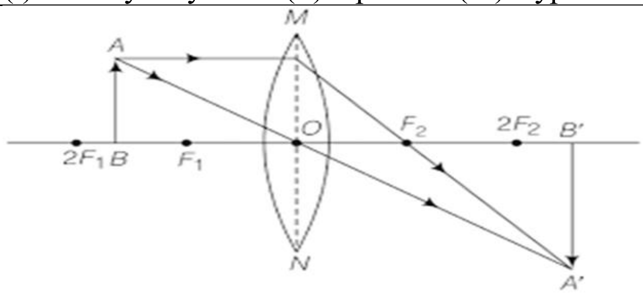
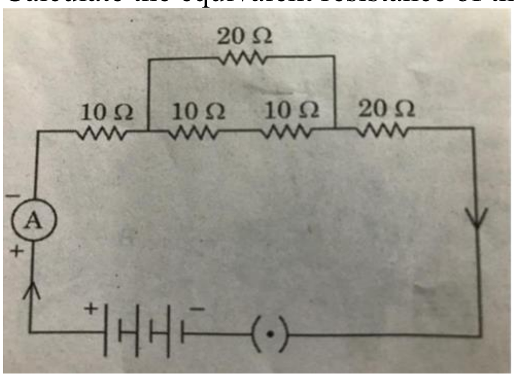
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true

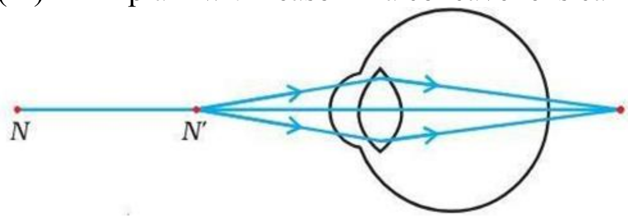
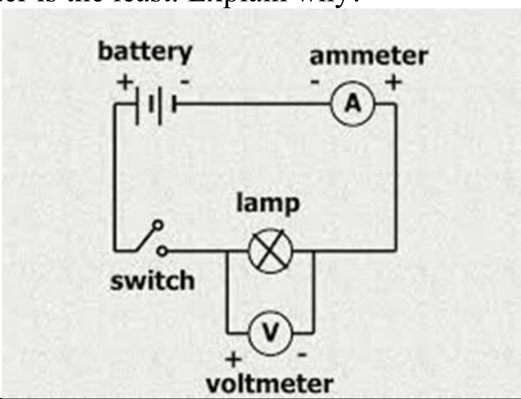
17 Assertion (A): ionic compounds conduct electricity in the solution as well as in solid state (1)
Reason (R):. the conduction of electricity through an ionic solution involves the movement of charged particles

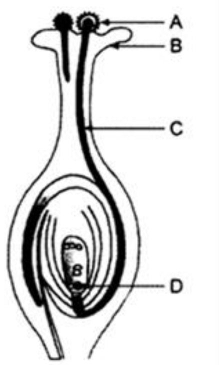
18 Assertion (A): The number of chromosomes in a cell and in a germ cell is not the same in any species. (1)
Reason (R): When 2 germ cells combine they restore the normal number of chromosomes in a species.

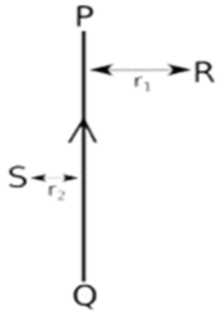
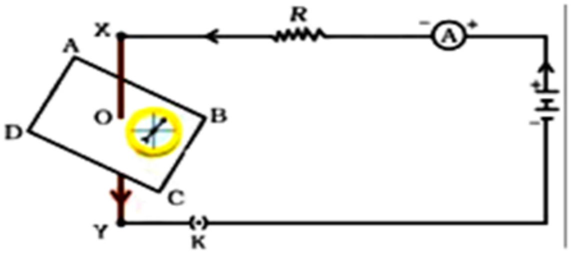
19 Assertion (A): When a concave mirror is held under water, its focal length will increase. (1)
Reason (R): The focal length of a concave mirror is independent of the medium in which it is placed.

20 Assertion (A): All organisms have multiple roles and interactions (1)

	Reason (R): Food webs are complex networks were organisms can occupy different trophic levels simultaneously leading to diverse ecological interaction.	
	Section-B Question No. 21 to 26 are very short answer questions	
21	Illustrate the formation of NaCl from sodium and chlorine atoms by the transfer of electron(s).	2
22	Differentiate between alveoli and nephron on the basis of the following points: 1.Structure and location 2.Function	2
23	**Attempt either option A or B.** A. Patients whose gallbladder are removed are recommended to eat less oily food. Why? OR B.Explain the role of the following enzymes in the process of digestion of food in humans: (i)Salivary amylase (ii)Pepsin (iii)Trypsin (iv)Lipase	2
24	 The above figure shows the formation of an image by a lens. Analyse the figure and answer the following questions. A. Is the type of lens converging or diverging? B. In which position of the object will the magnification be -1	2
25	Attempt either option A or B. A. Calculate the equivalent resistance of the following electric circuit  OR B. In what time 400 joules of heat is produced across a 16 ohm resistor at 80 V potential difference?	2
26	Create a food chain with more than 2 trophic levels that exists in the cabbage farm. If Humans occupy the last trophic level, then how would spraying pesticide affect the humans? Explain.	2
	Section-C Question No. 27 to 33 are short answer questions	

27	Give reason for the following: i)Metals are mostly used for making cooking utensils ii) Hydrogen gas does not evolve when certain metals except Mg & Mn react with nitric acid.	3
28	Attempt either option A or B. A. An element 'M' with electronic configuration 2 8 3 combines separately with Cl^{-} , SO_4^{-2} anions. (i)Write the chemical formulae of the compounds formed. (ii)What is the type of the chemical bond formed by element 'M' generally? Give reason. (iii) How will the electrical conductivity of the element 'M' differ from that of the compounds formed by'M'? OR B. A reddish-brown metal 'X', when heated in air, gives a black compound 'Y', which when heated in presence of H_2 gas gives 'X' back. 'X' is refined by the process of electrolysis. X' is used in electrical wiring. Identify 'X' and 'Y'. Draw a well-labelled diagram to represent the process of refining 'X'	3
29	With the help of a schematic diagram show the various pathways in the breakdown of glucose during cellular respiration.	3
30	The blood test of a person reveals that he has high levels of glucose in his blood due to the deficiency of a hormone. (i)Identify the hormone and name the gland which produces this hormone. (ii)Name the health hazard he is suffering from. (iii)Suggest a possible remedy to improve his health condition.	3
31	A. What is the fundamental difference between hypermetropia and myopia in terms of the optical experience of a person? B. The diagram below shows a special case of an eye defect. (i) What is the defect that is shown in the figure? (ii) State one cause for such a defect? (iii) Explain with reason if a concave lens can be used to correct the defect. 	3
32	A. Of all the components in the circuit shown below, the current flowing through the voltmeter is the least. Explain why? 	3

	B. A nichrome Wire has a resistance of 10 ohm. Find the resistance of another nichrome wire whose length is three times more and area of cross section 4 times more compared to the first wire.	
33	A student fixes a white sheet of paper on a drawing board. He places a bar magnet in the centre and sprinkles some iron filings uniformly around the bar magnet. Then he taps gently and observes that iron filings arrange themselves in a certain pattern. (a) Why do iron filings arrange themselves in a particular pattern? (b)Draw the pattern of magnetic field lines around the bar magnet. (c) State any two properties of magnetic field lines.	3
	Section-D Question No. 34 to 36 are long answer questions.	
34	Attempt either option A or B. A. Vedha heated Ethanol with a compound A in presence of a few drops of concentrated sulphuric acid and observed a sweet smelling compound B is formed. When B is treated with sodium hydroxide it gives back Ethanol and a compound C. (a) Identify A and C (b) Give one use each of compounds A and B. (c) Write the chemical equations for the above reactions. OR B. (a) What is the role of concentrated Sulphuric acid when it is heated with Ethanol at 443 K. Give the reaction involved. (b) Akash by mistake forgot to label the two test tubes containing Ethanol and Ethanoic acid. Suggest an experiment to identify the substances correctly. Illustrate the reactions with the help of chemical equations.	5
35	Attempt either option A or B. (a) A. Draw a diagram of human female reproductive system and label the part (i) that produces eggs. (ii) site of fertilization (iii) where zygote is implanted. (b) What happens to human egg when it is not fertilised? OR (a) List two reasons for the appearance of variations among the progeny formed by sexual reproduction.’ (b) Study the given diagram and answer the following questions. 	5

	(i) Name the part marked 'A' in the diagram. (ii) How does 'A' reach part 'B" (iii) State the importance of part 'C'. (iv) List any two post fertilisation changes in a flower.	
36	Attempt either option A or B. A.PQ is a current carrying conductor in the plane of the paper as shown in the figure below.  (i)Find the directions of the magnetic fields produced by it at points R and S? (ii) Given r1>r2 ; Where will the strength of the magnetic field be larger?Give reason. (iii) If the polarity of the battery connected to the wire is reversed, how would the direction of the magnetic field be changed? (iv) Explain the rule that is used to find the direction of the magnetic field for a straight current carrying conductor. OR B.(1) A student placed a compass needle near a conductor wire and passed electric current through the wire and found that the compass needle is deflected.  What difference will he find in the deflection of compass needle this time, when (a) Magnitude of electric current through the conductor is increased. (b) The compass needle is moved away from the conductor . (2) Draw the magnetic field lines across a 1. Solenoid 2. Circular loop	5
Section – E Question No. 37 to 39 are case-based/data -based questions		
37	In an electric circuit, electric fuse prevents overloading. Overloading occurs when live wire and the neutral wire come in direct contact with each other, causing the entry of an extra large current in the circuit. Overloading may also occur because of connecting many appliances to a single socket. The Earth wire which is thick having green insulation is connected to a metal plate deep in the earth near the house. This type of safety measure is used in appliances like electric press, toaster, table fan, refrigerator etc. The Earth wire gives low resistance conducting path for the electric current. In this way it protects us from severe electric shock.	4

	A. Write two reasons for overloading taking place in domestic electric circuit. B. Name two safety devices used in domestic electric circuit. C. How does earth wire provide protection? OR A fuse wire permits electric current upto 5 A in a circuit.To permit a current of 10 A through the circuit, should the new fuse wire (of same material) be thicker or thinner than the earlier one? Give reason for your answer.	
38	Paheli's science teacher was discussing the topic on sex determination in organisms. Different species use very different strategies for this. Some rely on environmental cues , others on the temperature to which fertilised eggs are subjected to. Some animals like snail can change their sex. However, in human beings, sex is genetically determined. A. Why is the pair of chromosomes in males called a mismatched pair? B. How many pairs of sex chromosomes are found in a normal cell of Human body? C. Mother is not responsible for the sex of the child. Justify your answer. OR With the help of a flow chart, show sex determination in human beings.	4
39	Cleaning agents are compounds that are used to remove stains, dust, unpleasant odors, dirt, and other contaminants from surfaces. Soaps and detergents are cleansing agents The most of dirt are oily in nature .Oil does not dissolve in water. The ionic-end of soap interacts with water while the carbon chain interacts with oil. The soap molecules, thus form structures called micelles, This forms an emulsion in water. The soap micelle thus helps in pulling out the dirt in water and we can wash our clothes clean. A. A student studies that a soap molecule has two ends, one of which is an ionic end and the other is the carbon chain. Which option explains the interaction of a soap molecule with oil? (a)ionic end of the soap interacts with the oil (b)the closest end of the soap interacts with the (c)carbon chain end of the soap interacts with the oil (d)ends of the soap randomly interact with the oil B. How do soaps differ from detergents? C. How are micelle formed? OR Oil does not dissolve in water. How do soap molecules bring oily dirt into water?	4